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PROJECT SYNOPSIS

ON

CYBERSECURITY THREAT DETECTION PLATFORM

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INTRODUCTION

Cybersecurity is becoming very important because most data and services are now available online. With the increase in internet usage, cyber threats such as unauthorized access, repeated failed login attempts, suspicious IP activity, and abnormal requests are also increasing.

The Cybersecurity Threat Detection Dashboard is a web-based application designed to monitor security activities and detect suspicious behaviour. The system will analyse security logs and generate alerts when a possible threat is found.

The dashboard will help the administrator view recent threats, security alerts, severity levels, and other important information in one place. This makes it easier to identify and understand security problems.

The frontend of the project will be developed using HTML, CSS, and JavaScript, the backend will use Python with Flask, and MySQL will be used to store security logs, users, threats, and alerts.

The main purpose of this project is to create a simple and useful system that can detect common cybersecurity threats and display them clearly through an interactive dashboard.

OBJECTIVE

The main objective of the Cybersecurity Threat Detection Dashboard is to develop a simple system that can monitor security activities, detect suspicious behaviour, and display important threat information on a dashboard.

The key objectives of the project are:

- To monitor different security activities and user actions.
- To detect suspicious login attempts and unauthorized access.
- To identify possible cyber threats using simple detection rules.
- To generate alerts whenever suspicious activity is found.
- To classify threats according to their severity, such as Low, Medium, High, or Critical.
- To store security logs, threat details, and alerts in the database.
- To display recent threats, alerts, and security information clearly on the dashboard.
- To help administrators understand and monitor security issues easily.
- To provide a simple and user-friendly way to manage cybersecurity information.

METHODOLOGIES

The Cybersecurity Threat Detection Dashboard will be developed step by step using a simple software development process.

1. Requirement Analysis: Identify the main features of the system such as login, security monitoring, threat detection, alerts, and dashboard.
2. Frontend Development: Use HTML, CSS, and JavaScript to design the login page and cybersecurity dashboard.
3. Backend Development: Use Python and Flask to handle system logic, user requests, and threat detection.
4. Database Management: Use MySQL to store user details, security logs, detected threats, and alerts.
5. Threat Detection: Analyse security activities and detect suspicious behaviour using predefined rules.
6. Alert Generation: Generate alerts when any suspicious or harmful activity is detected.
7. Dashboard Display: Show threats, alerts, severity levels, and security information in a simple and clear format.
8. Testing: Test the system with normal and suspicious activities to make sure all features work correctly.

HARDWARE REQUIREMENTS

The Cybersecurity Threat Detection Dashboard does not require any special hardware.

- Processor: Intel Core i3 or above
- RAM: Minimum 4 GB, 8 GB recommended
- Storage: Minimum 10 GB free space
- Display: Standard monitor or laptop screen
- Input Devices: Keyboard and mouse
- Internet Connection: Required for downloading tools and development resources

SOFTWARE REQUIREMENTS

The following software will be used to develop the Cybersecurity Threat Detection Dashboard:

- Operating System: Windows 10/11 or Linux
- Code Editor: Visual Studio Code
- Frontend: HTML, CSS, and JavaScript
- Backend: Python with Flask
- Database: MySQL
- Database Tool: MySQL Workbench
- Web Browser: Google Chrome or Microsoft Edge
- Chart Library: Chart.js for graphs and dashboard charts
- Version Control: Git and GitHub

These software tools will help in designing the interface, developing the backend, storing data, and testing the complete system.

OUTCOME

The expected outcome of the **Cybersecurity Threat Detection Dashboard** is to create a simple and useful system that can monitor security activities and identify possible cyber threats.

The system will be able to:

- Detect suspicious activities and failed login attempts.
- Generate alerts when a possible threat is found.
- Classify threats based on their severity.
- Store security logs and threat details in the database.
- Display recent threats and alerts on the dashboard.
- Show security information using charts and reports.
- Help administrators understand and monitor security problems easily.

The final system will provide a clear and user-friendly way to view and manage cybersecurity threats.

APPLICATIONS

The **Cybersecurity Threat Detection Platform** can be used in different applications where monitoring security activities is important. Some of its main applications are:

1. **Web Applications** – It can monitor login activities and detect repeated failed login attempts.
2. **Small Organizations** – It can help organizations keep track of suspicious activities in their systems.
3. **Server Monitoring** – Security logs from servers can be monitored to find unusual activities.
4. **Network Security** – It can help identify suspicious IP addresses and possible attacks.
5. **Educational Purpose** – The platform can be used by students to understand how security logs and threat detection systems work.
6. **Security Monitoring Dashboard** – It provides a simple dashboard where security events and detected threats can be viewed easily.

FUTURE SCOPE

In the future, the project can be improved by adding more security features. The possible improvements are:

1. **More Attack Detection** – Detection of attacks such as SQL Injection, Port Scanning and Denial-of-Service attacks can be added.
2. **Email Alerts** – The system can send an email when a serious threat is detected.
3. **Better Data Visualization** – More graphs and charts can be added to understand security data easily.
4. **Machine Learning** – Machine learning can be used in the future to identify unusual activities automatically.
5. **Real-Time Monitoring** – WebSocket or similar technology can be used for faster real-time threat updates.
6. **User Login System** – Authentication can be added so that only authorized users can access the dashboard.
7. **Deployment** – The platform can be deployed on a server so that it can be accessed and monitored remotely.

REFERENCE

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